

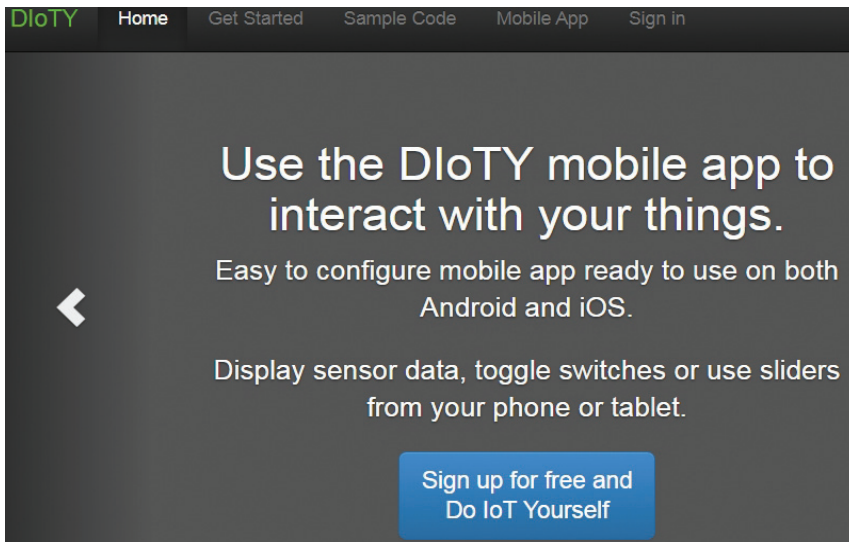


Fig. 5: Dashboard of DIoT

Inject a message here:

Topic: /kumargaurav.in@gmail.com/ Message:

Adapt your subscription to view only a subset of your messages:

Topic: /kumargaurav.in@gmail.com/ ☐ Keep history

(3:01:13 PM) /kumargaurav.in@gmail.com/: Signal Received

Fig. 6: Receiving data on the DIoT dashboard using MQTT

```
mqttclient.on_connect = on_connect
mqttclient.on_publish = on_publish
mqttclient.on_subscribe = on_subscribe
mqttclient.on_message = on_message
# Connect
mqttclient.username_pw_set('kumargaurav.in@gmail.com', '7881730a')
mqttclient.connect('myMQTT.dioty.co', 1883)
# Subscribing
mqttclient.subscribe('/kumargaurav.in@gmail.com/')
# Publishing a message
x=input('Message')
mqttclient.publish('/kumargaurav.in@gmail.com/', x)
# Loop; exit on error
CN = 0
while CN == 0:
    CN = mqttclient.loop()
    print("CN: " + str(CN))
```

After executing the Python script, the data is handled and processed by MQTT broker, as shown in Fig. 6.

Scope for research and development

Research scholars, academicians, and practitioners can carry out research and make innovative problem statements on IoT using MQTT brokers. A number of research statements and domains exist in which MQTT based implementations can be done including telemedicine, personal security gadgets, defence equipment, military devices, agribots for smart agriculture, and many others. **EFY**

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Dr Gaurav Kumar is managing director of Magma Research and Consultancy and is associated with various academic institutes, where he delivers expert lectures and conducts technical workshops on the latest technologies and tools

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